

Executive dysfunction in Parkinson's disease: From neurochemistry to circuits, genetics and neuroimaging

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ABSTRACT

Cognitive decline is one of the most significant non-motor symptoms of Parkinson's disease (PD), with executive dysfunction (EDF) being the most prominent characteristic of PD-associated cognitive deficits. Currently, lack of uniformity in the conceptualization and assessment scales for executive functions impedes the early and accurate diagnosis of EDF in PD. The neurobiological mechanisms of EDF in PD remain poorly understood. Moreover, the treatment of cognitive impairment in PD has progressed slowly and with limited efficacy. Thus, this review explores the characteristics and potential mechanisms of EDF in PD from multiple perspectives, including the concept of executive function, commonly used neuropsychological tests, neurobiochemistry, genetics, electroencephalographic activity and neuroimaging. The available evidence indicates that degeneration of the frontal-striatal circuit, along with mutations in the Catechol-O-methyltransferase (COMT) gene and Leucine-rich repeat kinase 2 (LRRK2) gene, may contribute to EDF in patients with PD. The increase in theta and delta waves, along with the decrease in alpha waves, offers potential biomarkers for the early identification and monitoring of EDF, as well as the development of dementia in patients with PD. The PD cognition-related pattern (PDCP) pattern may serve as a tool for monitoring and assessing cognitive function progression in these patients and is anticipated to become a biomarker for cognitive disorders associated with PD. The aim is to provide new insights for the early and precise diagnosis and treatment of EDF in PD.

1. Introduction

Cognitive impairment is six times more prevalent in patients with Parkinson's disease (PD) compared to healthy individuals. It represents one of the most significant non-motor manifestations of PD and is a crucial aspect of the disease's natural history. This impairment can substantially impact patients' quality of life and overall functioning. Moreover, it has been demonstrated to incur greater costs than

managing motor symptoms, even in the early stages of PD, making it a priority for both patients and caregivers (Aarsland et al., 2021). Executive dysfunction (EDF) is the most notable feature among cognitive impairments associated with PD, affecting approximately 30 % of patients (Dirnberger and Jahanshahi, 2013). These deficits can manifest early in the disease progression and involve a disruption in the coordination of a range of cognitive processes essential for executing complex, goal-oriented, and novel cognitive tasks (Parker et al., 2013). In this

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review, we discuss the concept of executive function and the heterogeneity of executive function scales. We also explore the characteristics of executive dysfunction in PD, drawing from studies on neurotransmitters, genetics, electroencephalographic activity, neuroimaging, and other related research. Cognitive dysfunction in PD may be identified at an early stage from diverse perspectives.

2. Definition of Executive Function (EF)

The renowned Russian neuropsychologist Alexander Luria first introduced the concept of ‘executive function’ within his theoretical model of brain function (Martínez et al., 2016). Pribram subsequently underscored the significant role of the frontal lobe in executive function, particularly in behavioral regulation (Luria et al., 1964). He posited that the prefrontal cortex serves as the ‘executive’ of the brain, playing a pivotal role in executive function and facilitating higher-order cognitive processes that enable flexible and adaptive behavior (Brody and Pribram, 1978). Shallice focused on the role of attention in behavioral control and proposed the ‘supervisory attentional system’ to regulate and enhance control over behavioral choices, especially in complex tasks (Norman and Shallice, 1986). Research by Posner has focused primarily on selection (Posner, 1966), allocation (Posner, 1969), and orientation of attention (Klein and Posner, 1974), as well as the interaction and independent effects of attention and vision (Posner, 1980). In 1990, leveraging advancements in brain imaging technology, neurophysiological mechanisms, and cognitive psychology experimental paradigms, Posner proposed the attention network system theory. This theory categorizes attention into three distinct networks: the alerting network, the orienting network, and the executive control network (Posner and Petersen, 1990). In 2012, Posner further enhanced the understanding of the attention system by emphasizing the independence of its functions, the integration of its neural basis, and the interactive effects among these networks (Petersen and Posner, 2012). Baddeley and Hitch significantly contributed to the theory of executive function by introducing the three-component model of working memory, which comprises the central executive system, the phonological loop, and the visuospatial sketchpad (Baddely et al., 1974). In 2000, Baddeley expanded this model by proposing the “episodic buffer” as its fourth component (Baddeley, 2000). In 2011, he further elaborated on the roles of the episodic buffer and the central executive system, examining these functions from both theoretical and neurobiological perspectives (Baddeley et al., 2011; Baddeley, 2012). Baddeley’s theoretical model has endured, demonstrating its relevance in understanding attention processing resources and temporary storage, as well as the essential distinctions between temporary linguistic and visuospatial information. However, research on the central executive and the episodic buffer has been relatively limited. The future integration of various theoretical models and interdisciplinary exploration appears to be highly promising (Hitch et al., 2024). Nevertheless, a precise and widely accepted definition of executive function has yet to be established. We have synthesized the common theoretical models of executive function in Table 1.

3. Neuropsychological tests commonly used for executive function

Commonly used working memory assessments include the Digit Span Test (DST), N-back tests, etc. The Stroop Test, Go/No-Go Test and Simon Task are primarily used to evaluate inhibitory control capabilities. Tests such as Wisconsin Card Sorting Test (WCST), verbal fluency, category fluency and Trail Making Test (TMT) are used to assess cognitive flexibility (Litvan et al., 2012)(Table 2).

The Stroop Color-Word Test (SCWT), first introduced by Stroop in 1935, is commonly used in neuropsychology and neuropsychiatry to assess cognitive inhibitory control and selective attention. Despite its long history and widespread application, the exact neural mechanisms underlying the SCWT remain incompletely understood. Neuroimaging

Table 1
Common Theoretical Models of Executive Function.

Model	Year	Author
Luria model	1973	Luria
Working memory model	1974	Baddeley & Hitch
Duncan’s Theory of Goal Neglect	1986	Duncan
Supervisory attentional system (SAS)	1986	Norman & Shallice
The Attention System	1990	Posner
Goldman-Rakic’s working memory model	1992	Goldman-Rakic
Damasio’s somatic marker hypothesis	1995	Damasio
Borkowski and Burke’s Information-processing	1995	Borkowski & Burke
Lezak’s conceptual model	1995	Lezak
Self-regulatory model	1997	Barkley
Problem-solving model	1997	Zelazo, Carter, Reznick & Frye
Fuster’s Theory of Cross-Temporal Organization	1997	Fuster
Executive Control System Model	2002	Anderson
Banich’s “Cascade of control” model	2009	Banich

Table 2
Common Executive Function Neuropsychological Tests.

Commonly Used Executive Function Tests	Involved Executive Function Processes
Wisconsin Card Sorting Test (WCST, various versions) and other card sorting tests	Cognitive flexibility, problem-solving, inhibition and switching, sustained selective attention
Stroop Test	Conflict resolution, response inhibition, selective attention
Verbal Fluency Test (VFT)	Cognitive flexibility, inter-cluster switching, monitoring, strategic search, attention maintenance, response inhibition, response generation
Semantic Fluency Test (SFT)	Semantic knowledge
Phonemic Fluency Test (PFT)	Strategic search
Digit Span Test (DST)	
Digit Span Backward (DST-B)	Strategic search
Digit Span Forward (DST-F)	Short-term memory
Tower of London (ToL) and other tower tests	Planning, inhibition, rule application
Trail Making Test (TMT)	
Trail Making Test A (TMT-A)	Perceptual processing, attention
Trail Making Test B (TMT-B)	Response inhibition, attention switching, cognitive flexibility

studies of patients and healthy controls indicate that the Anterior Cingulate Cortex (ACC) and Dorsolateral Prefrontal Cortex (DLPFC) are key brain regions involved in the cognitive control necessary for Stroop task performance (Henik and Salo, 2004). MacDonald et al. suggest that the DLPFC is associated with implementing control to overcome conflict through top-down mechanisms, while the ACC is linked to conflict detection and monitoring functions. Although prefrontal functions are crucial, the Stroop task is mediated by a more broadly distributed neural network (MacDonald 3rd et al., 2000).

The Digit Span Test comprises two assessments: the Digit Span Forward (DST—F) and the Digit Span Backward (DST—B). DST-F involves the repetition of verbal information without processing, assessing the patient’s short-term memory and relying on the Ventrolateral Prefrontal Cortex (VLPFC). On the other hand, DST-B requires patients to process information obtained without external auditory prompts and then reproduce it, primarily testing working memory capabilities that are dependent on the DLPFC (Diamond, 2013; Conklin et al., 2000).

The Trail Making Test (TMT) has two parts: Trail Making Test A (TMT-A) and Trail Making Test B (TMT—B). The TMT assesses various cognitive functions, including visual processing (recognition and filtering of letters and numbers), cognitive flexibility, and attention switching. TMT-A mainly assesses the participant’s perceptual processing and attention, while TMT-B primarily evaluates response inhibition,

attention switching, and cognitive flexibility. Recent researches indicate that the TMT, in conjunction with other tests, may help identify early signs of neurodegenerative conditions like Alzheimer's disease and Huntington's disease (Catalan et al., 2018; O'Rourke et al., 2011).

The Verbal Fluency Test is utilized primarily to evaluate cognitive flexibility and consists of two main tests: the Category Fluency Test (CFT), also known as the Semantic Fluency Test (SFT), and the Word Fluency Test (WFT), also known as the Phonemic Fluency Test (PFT) (Robinson et al., 2021). According to the neuroimaging studies, PFT is primarily reliant on the frontal cortex associated with strategic searching abilities, while SFT is more dependent on the temporal cortex and related to semantic knowledge (Kavé, 2005; Tanida and Saito, 2022). Increasing clinical researches suggest that SFT can identify preclinical stages of Alzheimer's disease, with a decline in scores indicating disease progression. It has also been proposed that impairments in SFT are more pronounced in patients with frontotemporal dementia (Fuster, 2019).

The Prefrontal Cortex (PFC) has been proven to be a crucial structure for executive function (Friedman and Robbins, 2022), with damage to the PFC associated with deficits in impulse control, planning strategies and inhibitory behaviors required for task completion (Callo et al., 2012). A meta-analysis conducted by Yuan et al. with 3272 participants revealed a positive correlation between PFC volume and executive function performance. The study found that individuals with larger PFC volumes tended to perform better on executive function tests, providing evidence in support of the 'bigger is better' hypothesis (Yuan and Raz, 2014).

However, increasing researches suggest that EDF in PD is not limited to specific cortical deficits but may also be related to neurotransmitter transmission disorders, changes in subcortical structures and regional connections, as well as genetics (Mirpour et al., 2022).

4. Association between executive function and neurotransmitter systems in Parkinson's disease

4.1. Dopaminergic system and executive function

Dopaminergic neurons, which are diverse in both anatomy and function, are found in various regions such as the diencephalon, midbrain and olfactory bulb, primarily constituting the nigrostriatal system, the mesolimbic system and the mesocortical system (Fig. 1). The nigrostriatal system, originating from the substantia nigra pars compacta (A9 area) and extending fibers to the caudate-putamen complex

(dorsal striatum), is crucial for regulating voluntary movements (Fig. 1). It is involved in the frontal-striatal neural network, thus participating in the processes regulated by the frontal-striatal pathway such as motivation, working memory and emotion (Fig. 1). In individuals with PD, there is a varying degree of loss of nigrostriatal dopaminergic neurons, with an early reduction in dopamine levels primarily in the dorsal striatum and widespread decreases in dopamine neurotransmission as the disease progresses (Vaillancourt et al., 2013). The mesocortical pathway, which connects the VTA with the neocortex, especially the prefrontal cortex, insula and cingulate cortex, is essential for the maintenance of normal cognitive functions of the dorsolateral prefrontal cortex and is thought to be related to motivation and emotional responses. The mesolimbic system, originating from the VTA and projecting to the limbic system via structures like the nucleus accumbens, amygdala and hippocampus, as well as reaching the medial prefrontal cortex. While traditionally considered a 'reward' pathway, this concept is not universally accepted (Halliday et al., 2014).

The 'dopamine overdose hypothesis' suggests that following dopamine replacement therapy, PD patients exhibit worsened ability of reverse learning but improved performance in attention switching (Kehagia et al., 2010). This phenomenon is mainly related to the spatiotemporal process of dopamine loss in PD patients. Initially, there is a reduction in pigmented cells in the ventral portion of the substantia nigra (SN) followed by the dorsal portion. The ventral SN projects to the dorsal caudate nucleus (where dopamine reduction occurs early in the disease), while the dorsal SN projects to the ventral striatum. The ventral striatum also receives dopamine projections from the Ventral tegmental area (VTA); hence, there is no reduction in dopamine in the ventral striatum in the early stages of the disease. However, treatment with dopaminergic drugs can lead to overstimulation of the ventral striatum, resulting in a decline in the ability of reverse learning (Vaillancourt et al., 2013). Evidence indicates that this overdose effect may also occur in the prefrontal cortex, leading to deficits in attention switching, working memory and planning (Kehagia et al., 2010). Additionally, the 'dual syndrome hypothesis' posits that changes in dopaminergic transmission in the frontal-striatal network can contribute to executive dysfunction in PD patients, primarily characterized by deficits in planning, working memory, response inhibition and attention control (Kehagia et al., 2013).

4.2. Noradrenergic system – attention, cognitive flexibility

Noradrenergic neurons originate from the A1 to A7 groups of neurons that spread from the lateral brainstem to the medulla oblongata (Guyenet, 1991), with the A6 area (primarily within the locus coeruleus, dorsally to the pons, lateral to the fourth ventricle, and medial to the superior cerebellar peduncle) innervating most regions of the brain, including the substantia nigra and thalamus, the cholinergic nucleus basalis (NBM), limbic areas, and the cerebral cortex (Fig. 2). This innervation forms the locus coeruleus-nigral pathway, locus coeruleus-thalamic pathway, locus coeruleus-cholinergic nucleus basalis pathway, locus coeruleus-limbic cortical pathway, and locus coeruleus-cortical pathway (Halliday et al., 2014). The release of norepinephrine by the locus coeruleus is now recognized as playing a role in multiple cognitive functions, such as attention, behavioral flexibility, working memory, and long-term memory processes (Borodovitsyna et al., 2017).

Norepinephrine, along with the locus coeruleus-cortical projection, is closely related to cognitive flexibility and the attentional switching. The release of norepinephrine in the prefrontal cortex enhances the neuronal response to different incoming stimuli, which can subsequently facilitate cognitive flexibility (Robbins and Arnsten, 2009). Damage to the noradrenergic system can lead to deficits in cognitive flexibility, which can be ameliorated with noradrenergic agonists, as discussed by Logue et al., and this phenomenon is likely related to the closely associated medial prefrontal cortex (Logue and Gould, 2014). Notably, damage to the noradrenergic pathway is an early event in the

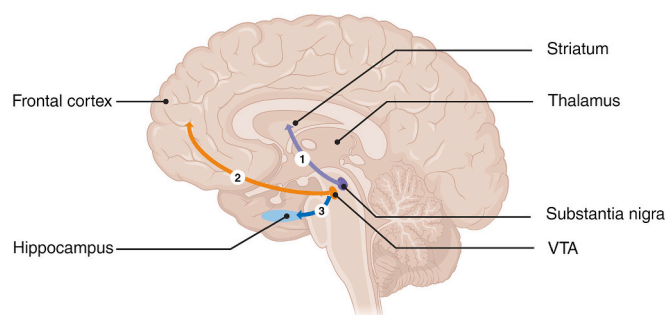


Fig. 1. Dopaminergic System and Executive Function.

The dopaminergic pathways: (1) The nigrostriatal system, originating from the substantia nigra pars compacta and extending fibers to the caudate-putamen complex (dorsal striatum); (2) The mesocortical pathway, which connects the VTA with the neocortex, especially the prefrontal cortex, insula and cingulate cortex; (3) The mesolimbic system, originating from the VTA and projecting to the limbic system; (4) Tuberoinfundibular, from the hypothalamus to the pituitary gland (not shown). The 'dual syndrome hypothesis' and the 'dopamine overdose hypothesis' posit that changes in dopaminergic transmission in the frontal-striatal network can contribute to executive dysfunction in PD patients (created with BioRender.com).

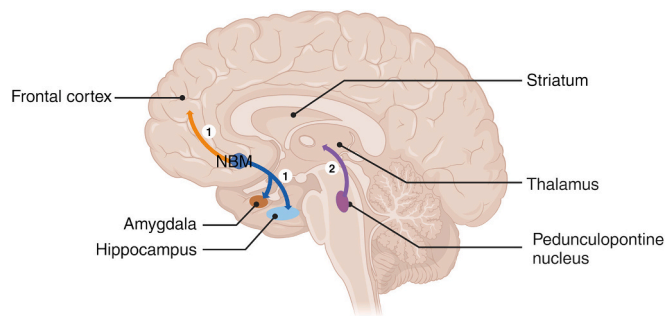


Fig. 2. Noradrenergic System – Attention, Cognitive Flexibility.

The noradrenergic pathways: (1) locus coeruleus-thalamic pathway; (2) locus coeruleus-cholinergic nucleus basalis pathway; (3) locus coeruleus-limbic cortical pathway; (4) locus coeruleus-cortical pathway; (5) the locus coeruleus-nigral pathway (not shown). Norepinephrine, along with the locus coeruleus-cortical projection, is closely related to cognitive flexibility and the attentional switching. N. acc: nucleus accumbens; NBM: the cholinergic nucleus basalis. (created with [BioRender.com](#)).

pathological process of Parkinson's disease (Frouni et al., 2022), occurring during its prodromal phase (Braak II). In early PD patients, attention deficits can be exacerbated due to dysfunction of noradrenergic neurons, and further neuroimaging studies have found that changes in attention and reaction time are regulated by the locus coeruleus-modulated frontoparietal attention network (Brandão et al., 2020). Reduced availability of the noradrenaline transporter in the brain is linked to cognitive decline and orthostatic hypotension in PD. This association may be attributed to brain hypoperfusion resulting from orthostatic hypotension, which can negatively impact cognitive function. The use of noradrenergic-enhancing medications is recommended to treat orthostatic hypotension and potentially improve cognitive function in PD patients (Longardner et al., 2020; McDonald et al., 2016; Sinn and Gibbons, 2016).

4.3. Serotonergic system – sleep, depression, anxiety

Serotonergic neurons are primarily found in specific areas of the brain, including the midbrain raphe nuclei, upper pons, and median raphe nucleus of the medulla oblongata, organized into nine groups (Halliday et al., 2014; Fox et al., 2010). Normally, the median raphe nucleus densely innervates the basal ganglia and likewise projects to the frontal lobe, limbic system and the hypothalamus (Fig. 3). The serotonergic system regulates essential physiological functions like body temperature, sleep, appetite, pain and motor activity, while also influencing higher brain functions such as cognition and emotion. High-

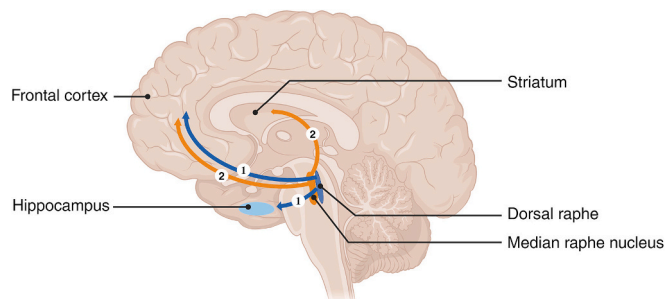


Fig. 3. Serotonergic System – Sleep, Depression, Anxiety.

The serotonergic pathways: (1) the dorsal raphe projects to the striatum and cortex, (2) while the median raphe projects to limbic regions and cortex. The relationship between the degeneration of serotonergic neurons and cognitive impairment is not fully understood in PD patients, but it is linked to other non-motor symptoms like sleep disturbances, depression, anxiety and hallucination (created with [BioRender.com](#)).

density serotonergic projections to the hippocampus and prefrontal cortex indicate anatomical and neurochemical links between the serotonergic system and the brain regions crucial for learning and memory (Švob Štrac et al., 2016). Clarke et al. selectively depleted serotonin (5-HT) in the prefrontal cortex of marmosets, which led to cognitive flexibility deficits, thereby confirming the critical role of 5-HT in cognitive behavioral flexibility (Clarke et al., 2007). Zheng et al. demonstrated that the selective serotonin reuptake inhibitor citalopram can enhance behavioral impulse control, specifically in terms of behavioral inhibition, possibly through enhanced activation of the inferior frontal cortex. This treatment is most effective for patients with more severe symptoms and contributes to a deeper understanding of how the serotonin system influences cognitive and behavioral regulation (Ye et al., 2014). Espay et al. discovered that the serotonin 2 A receptor inverse agonist pimavanserin demonstrated a notable antipsychotic effect in PD psychosis patients with cognitive impairment, particularly in hallucinations and delusions. They suggested that combining pimavanserin with cognitive enhancement drugs may enhance its efficacy (Espay et al., 2018). However, the researchers did not find evidence to support the improvement of cognitive dysfunction in Parkinson's disease patients. In PD patients, the relationship between the degeneration of serotonergic neurons and cognitive impairment is not fully understood, but it is linked to other non-motor symptoms like sleep disturbances, depression, anxiety and hallucination (Longardner et al., 2020; Maillet et al., 2016).

4.4. Cholinergic system – visuospatial and memory

Cholinergic neurons in the brain are primarily found in three groups: interneurons in the striatum, basal forebrain neurons (e.g., Ch4 neurons of the nucleus basalis of Meynert), and brain stem cholinergic neurons (Ch5 neurons of the pedunculopontine nucleus) (Halliday et al., 2014; Maurer and Williams, 2017). These neurons release acetylcholine (ACh) that projects to a wide range of cortical and subcortical areas (Fig. 4). Previous physiological, pathological, pharmacological and genetic studies have extensively explored the role of acetylcholine in cognitive functions like attention and memory. Research by Maurer et al. has further discovered changes in cholinergic signaling in patients with attention and cognitive control disorders (Maurer and Williams, 2017). Newman et al. believe that acetylcholine is crucial in various cognitive domains, including working memory, attention, episodic memory, encoding and spatial memory processing (Newman et al., 2012). The 'dual syndrome hypothesis' proposed by Kehagia suggests that the cholinergic neurotransmitter pathways in the posterior cortical regions of PD patients are closely related to visuospatial functions and memory functions (Fig. 4); this theory supporting the clinical use of acetylcholinesterase inhibitors and other drugs to improve patients' cognitive functions (Nombela et al., 2014). It is noteworthy that in Parkinson's

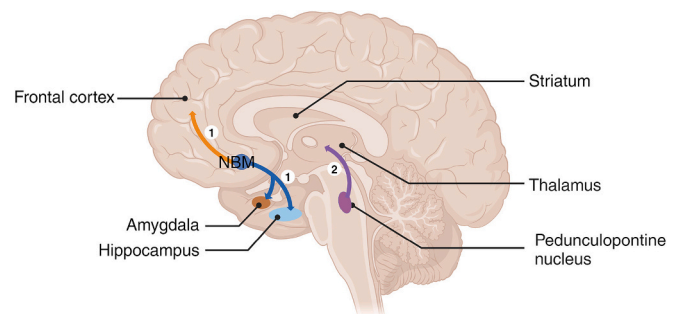


Fig. 4. Cholinergic System – Visuospatial and Memory.

The cholinergic pathways: (1) from the pedunculopontine nucleus to the thalamus; (2) from the basal nucleus of Meynert to the limbic regions and cortex. The cholinergic neurotransmitter pathways in the posterior cortical regions of PD patients are closely related to visuospatial functions and memory functions (created with [BioRender.com](#)).

Disease with dementia (PD-D) patients, the loss of cholinergic fibers in the brain is more evident than the loss of cholinergic neurons (Alexandris et al., 2020); this degeneration of acetylcholine innervation is associated with cognitive decline, and the loss of basal forebrain cholinergic projections to the hippocampus is related to memory deficits and the transition to PD-D. In PD-D patients, there is more than a 50 % degeneration of neurons in the NBM, and the posterior-anterior gradient of cortical cholinergic loss is evident. This may be due to a significant cell loss in the Ch4 neuron group affecting cholinergic innervation to the occipito-parietal cortical regions (Liu et al., 2015). Regression analysis by Bohnen et al. indicates that the denervation of dopaminergic neurons in the caudate nucleus and the degeneration of cortical cholinergic have a ‘synergistic effect’ on cognitive impairments in PD (Bohnen et al., 2015).

5. Genetic components of executive dysfunction in PD

Genetic mutations have been identified as a contributing factor to cognitive impairments in patients with PD, with carriers of pathogenic genes experiencing more severe and faster progressing cognitive impairments compared to non-carriers. In the following section, we provide a summary of our findings (Table 3).

5.1. Catechol-O-methyltransferase (COMT) gene

COMT plays a crucial role in regulating synaptic dopamine levels, especially in the frontal cortical area (Gonzalez-Latapi et al., 2021). A substitution of methionine for valine at position 158 of COMT (Val → Met) leads to a 40 % reduction in COMT activity and higher dopamine levels in the prefrontal cortex (Chen et al., 2004), whereas Val carriers have higher COMT enzymatic activity, leading to lower dopamine levels in the frontal regions (Hirano, 2021). In PD patients, carriers of the high-activity COMT haplotype show cognitive decline and a reduction in grey matter volume (Sampedro et al., 2020a; Lin et al., 2018). Early PD patients with the Met allele have shown impaired performance in planning tasks that rely on frontal cortical regions (Gonzalez-Latapi et al., 2021). This result has also been confirmed in the CamPaIGN study, where executive functions improved in COMT Met/Met patients over the five subsequent years of disease progression, which might be explained by the ‘Dopamine overdose hypothesis.’ (Collins and Williams-Gray, 2016).

5.2. Glucocerebrosidase (GBA) gene

The GBA gene, situated on chromosome 1’s short arm, encodes the lysosomal enzyme β-glucocerebrosidase (GCase) and is identified as an independent risk factor for cognitive impairment in individuals with PD (Alcalay et al., 2012). PD patients with GBA gene mutations experience more rapid cognitive decline, have a significantly increased risk of dementia (Halliday et al., 2014), exhibit more severe motor symptoms and have a shorter lifespan compared to patients with idiopathic PD (Gonzalez-Latapi et al., 2021; Hirano, 2021). A multicenter cross-sectional study found that PD patients with the GBA gene E326K mutation had a higher incidence of dementia and performed worse in working memory/executive functions and visuospatial abilities (Mata et al., 2016). Research has also shown that low scores in verbal fluency

in PD-GBA patients are associated with reduced white matter in inter-hemispheric, frontal and parahippocampal regions (Barkhuizen et al., 2016). It has been proposed that the anatomical features of PD-GBA patients can be characterized by the PD cognition-related pattern (PDCP), marked by decreased metabolism in frontal and parietal association areas, while the cerebellar vermis and dentate nucleus exhibit relatively high metabolism (Hirano, 2021). Furthermore, mutations resulting in the loss of GBA activity increase the accumulation of glucosylceramide within cells, leading to reduced lysosomal degradation, subsequent accumulation of α-synuclein, and impaired clearance, affecting normal neuronal activity and causing motor and cognitive disorders (Lin and Farrer, 2014).

5.3. Leucine-rich repeat kinase 2 (LRRK2) Gene

The LRRK2 gene is the most common autosomal dominant pathogenic gene for PD, located on chromosome 12p11.2-p13.1 and containing 51 exons (Paisán-Ruiz et al., 2004). It plays a crucial role in various neuronal functions, such as neurogenesis, axonal growth, synaptic function, mitochondrial dynamics, transport of mitochondria and autophagy. LRRK2 is highly expressed in brain regions with dopaminergic innervation, such as the striatum, hippocampus, cortex, and cerebellum (Sweet et al., 2015), suggesting a direct correlation between the LRRK2 gene and motor and cognitive impairments in PD patients. G2019S is the most common mutation in the LRRK2 gene, and multiple clinical studies have found that carriers of the G2019S mutation among first-degree relatives of Ashkenazi PD patients are at high risk of developing PD. These individuals also exhibit reduced abilities related to attentional allocation, processing speed, interference resolution, and response inhibition, as assessed by the Stroop interference task (Thaler et al., 2013; Thaler et al., 2012) and these deficits correlate with brain regions such as the right inferior parietal lobule, right precuneus and right fusiform gyrus (Thaler et al., 2012). However, working memory assessed by the N-back task remains relatively intact in these carriers (Thaler et al., 2016). On the other hand, carriers of the R1441C mutation, second in prevalence to G2019S, do not show significant cognitive performance differences compared to individuals with idiopathic PD, as demonstrated by Estanga et al. (Estanga et al., 2014).

5.4. Microtubule-associated protein tau (MAPT) Gene

The MAPT gene is involved in the assembly and stabilization of microtubules. The MAPT H1/H1 genotype has been identified as a risk factor for tauopathy, with this haplotype increasing the expression of tau, which aggregates to form neurofibrillary tangles and filamentous pathological inclusion (Gonzalez-Latapi et al., 2021; Paul et al., 2016). Genome-Wide Association Studies (GWAS) have linked the MAPT H1 haplotype to a higher risk of PD and a faster cognitive decline in PD patients (Wise and Alcalay, 2022). The UK CamPaIGN cohort study found that carriers of the MAPT H1/H1 genotype showed a faster decline in Mini-Mental State Examination (MMSE) scores and progressed to dementia more rapidly. A 10-year follow-up study support the association between the H1/H1 genotype and dementia progression, although its impact lessened with disease duration, primarily affecting early cognitive decline (Gonzalez-Latapi et al., 2021). In a functional MRI study, the performance of early-stage PD patients with the MAPT H1/H1 genotype in memory and visuospatial tasks was shown to correlate with the functionality of the temporal and parietal regions, but not with the frontal-striatal pathway. This suggests that the MAPT H1/H1 effect specifically influences posterior cortical circuitry and may play a significant role in dementia development (Collins and Williams-Gray, 2016).

5.5. α-synuclein gene polymorphism (SNCA) Gene

Located on chromosome 4q21–q23, the SNCA gene consists of 140

Table 3
Genetic Influences on Cognitive Functions in Parkinson’s Disease.

Gene	Cognitive Function Changes in PD
COMT Val158Met	Executive Function
GBA	Dementia
APOEε4	Dementia
LRRK2	Executive Function
MAPT H1/H1	Memory and Visuospatial Function Impairment; Dementia
SNCA	Early Cognitive Impairment
DRD2/SLC6A3/BDNF	Executive Function

amino acids that encode α -synuclein, an autosomal dominant gene associated with PD (Winslow et al., 2010). The relationship between SNCA and PD-associated cognition is not yet consistent across studies. Longitudinal studies have indicated that carriers of the longer Rep1 alleles in the SNCA promoter region have a higher risk of developing executive dysfunction in early-stage PD patients (Tan et al., 2022). Antonini et al. suggest that triplication of the SNCA gene is associated with early-onset and rapidly progressing PD along with varying degrees of cognitive impairment (Antonini et al., 2012). Additionally, Koros et al. found that PD patients with the A53T point mutation experience cognitive challenges in verbal fluency, working memory, and visuospatial functions with a higher prevalence of prodromal symptoms such as olfactory dysfunction (Koros et al., 2018). However, in a large-sample multicenter study, the rs356219 polymorphism was not associated with any dimension of neuropsychological test scores in PD patients (Mata et al., 2014).

5.6. Other genetic factors

The Apolipoprotein E gene (APOE) ϵ 4 allele is associated with an increased burden of amyloid plaques. Cross-sectional studies have shown an overexpression of APOE ϵ 4 carriers among patients with PD-D, while longitudinal studies suggest that both APOE ϵ 2 and APOE ϵ 4 are related to faster cognitive decline and an increased risk of dementia in PD (Delgado-Alvarado et al., 2016). The dopamine receptor type 2 (DRD2) gene and the SLC6A3 gene, which are involved in dopamine transport, are closely associated with executive function in PD patients; DRD2 (rs2283265) genotypes and allele frequencies are linked with the risk of dementia (Zapala et al., 2022). The interaction between levodopa and DRD2 gene polymorphisms affects PD executive function in a task-dependent ‘inverted-U’ manner (Kwak et al., 2013). Dopamine transporters, involved in the clearance of dopamine from the striatal synaptic cleft, have a significant impact on frontal-striatal activation and performance in PD patients, with the SLC6A3/DAT1 genotype containing variable numbers of tandem repeats (VNTR), primarily comprising of 10-repeat and 9-repeat alleles; patients with the 9R allele (homozygous or heterozygous) perform worse on executive function and working memory and exhibit less activation compared to homozygous 10R allele carriers (Habak et al., 2014). Several single nucleotide polymorphisms (SNPs) in the CLU gene have been identified as susceptibility loci for Alzheimer’s disease. Newly diagnosed PD patients with the high-risk CLUSTERIN genotype may experience lower cognitive scores on memory and executive function tests, along with more significant cognitive decline and cortical thinning over a 5-year follow-up (Sampedro et al., 2020b). The BDNF gene is a risk factor for cognitive impairment in PD, and compared to homozygous BDNF-Val allele carriers (Yin et al., 2019), PD patients may benefit from the BDNF-Met allele in terms of cognitive flexibility and set-shifting (van der Kolk et al., 2015).

6. Relationship between electroencephalogram and cognitive impairment in Parkinson’s disease

Preliminary studies have indicated that electroencephalograms (EEGs) serve as predictive biomarkers for cognitive decline in PD patients (Aarsland et al., 2021). Compared with PD patients with normal cognitive function, those with PD-MCI may demonstrate increased theta wave activity in the frontal region, along with decreased alpha wave activity in the right temporal region (Kalia, 2018). Furthermore, the EEG activity observed in patients with PD-D is characterized by a slower rhythm, with increased power in the delta (1–4 Hz) and theta (4–8 Hz) frequency bands (Delgado-Alvarado et al., 2016).

A research led by Caviness investigated the cognitive processes of PD patients across various stages and concluded that delta wave power is significantly higher in the PD-D group than in the PD-MCI group (Caviness et al., 2016a). This alteration is associated with an increase in the level of cortical phosphorylated alpha-synuclein (Caviness et al.,

2016b). Additionally, there is a downward trend in the alpha wave power. Furthermore, the increase in theta band activity is also linked to cognitive decline, particularly in the early stages, which may indicate impaired memory function. The observed increases in theta and delta waves, alongside the decrease in alpha waves, represent frequency bands that are closely associated with cognitive impairment in Parkinson’s disease, as indicated by current quantitative electroencephalography (qEEG) studies. These changes in frequency bands may serve as potential biomarkers for the early identification and monitoring of cognitive function in PD patients (Geraedts et al., 2018; Caviness et al., 2015; Utianski et al., 2016).

Research conducted by Narayanan Labs provides substantial insights into the nature of low-frequency frontal rhythms, indicating that cognitive impairment in PD patients is linked to decreased delta and theta activity. These findings may facilitate the development of new biomarkers and targeted therapies for cognitive symptoms associated with PD (Singh et al., 2023). Initial studies revealed that low background rhythm frequencies (below 8.5 Hz) are significantly correlated with the risk of developing PD-D, with a risk ratio of 13. This observation has led to the hypothesis that qEEG biomarkers could play a crucial role in the early identification and intervention of PD-D (Klassen et al., 2011). Further investigations utilizing machine learning algorithms demonstrated that PD-related activities linked to cognitive function can occur synchronously across various spectral rhythms. Notably, the main oscillations captured by central-parietal electrodes revealed that theta-alpha (7–13 Hz) oscillation frequencies and beta oscillation amplitudes vary with cognitive function. Additionally, shifts in the spectral peaks of theta-alpha EEG rhythms in posterior cortical areas may serve as predictors for the onset of dementia in individuals with PD (Anjum et al., 2024).

Cavanagh and colleagues reported that during the execution of cognitive control tasks, theta and alpha wave activities in patients with PD were significantly lower than those observed in the control group (Singh et al., 2018). This finding reveals a direct relationship between these wave activities and cognitive dysfunction. In the same year, the team also determined that the mid-frontal delta/theta rhythms in PD patients exhibiting cognitive dysfunction were significantly reduced. This discovery provided a new biomarker for cognitive impairment associated with PD and established a theoretical foundation for future treatment strategies (Singh et al., 2023). Furthermore, Cavanagh and colleagues focused on medial frontal low-frequency rhythms (~ 4 Hz) in PD patients, which were significantly correlated with executive function in PD patients (Singh et al., 2021; Narayanan et al., 2024). This correlation may be attributed to decreased 4 Hz EEG activity, which affects the functional coupling between neurons in the mid-frontal cortex, thereby impacting temporal processing and executive function (Parker et al., 2015).

Although EEG is a non-invasive and convenient examination method, it has limitations in accurately localizing neuroanatomical impairments related to executive function in patients with PD. Consequently, this review further explores the neuroimaging characteristics associated with executive function impairment in PD patients.

7. Imaging manifestations of executive dysfunction in Parkinson’s disease (Table 4)

7.1. Structural Magnetic Resonance Imaging (sMRI)

sMRI provides information about the shape, size and integrity of brain structures. With the main approaches being voxel-based morphometry (VBM) for measuring changes in whole-brain grey matter density and volume, and surface-based morphometry (SBM) for morphological measurements of cortical surface area or cortical thickness (Balázová et al., 2019).

Numerous studies suggest that reductions in cortical volume in PD patients correspond with specific cognitive deficits. Mak et al. found that

Table 4

Summary of main MRI studies on neuroanatomical correlates of Executive dysfunctions in PD.

Study	Participants (mean age)	Assessment of executive function	MRI analysis	Correlation with brain structure
sMRI				
Xia, et al. (Xia et al., 2013)	32 PD (70.0) 25 HC (66.9)	the scores of visuospatial/executive function items in MoCA	VBM	caudate, the middle temporal gyri, the left precuneus and the cerebellum
Mak, et al. (Mak et al., 2014b)	23 PD-MCI (69.0) 67 PD-N (63.5)	FAB, the 10-point clock drawing test	VBM, SBM	Insular, left middle temporal gyrus, the superior frontal gyrus
Lee, et al. (Lee et al., 2014)	36 PD-MCI Non-converters (73.2) 15 PD-MCI Converters (70.6)	phonemic fluency, semantic fluency, word score in Stroop test and color-word score in the Stroop test	VBM	superior frontal gyrus, middle frontal gyrus
Filoteo, et al. (Filoteo et al., 2014)	51 PD (67.7) 39 HCs (64.8)	the Stroop Color-Word test, the Letter Fluency test	ROI	frontostriatal volumes
Koshimori, et al. (Koshimori et al., 2015)	26 PD (70.5) 15 HCs (67.1)	Visual Verbal Test total number of shifts, DKEFS Color-Word Interference (Stroop task), DKEFS Verbal Fluency	ROI, DTI	left dorsolateral superior frontal gyrus regions.
Study	Participants (mean age)	Assessment of executive function	MRI analysis	Correlation with brain structure
Gao, et al. (Gao et al., 2017)	23 PD-MCI (65.1) 23 PD-N (62.4) 21 HCs (63.8)	MoCA executive function domain	VBM	frontal lobe
Foo, et al. (Foo et al., 2017)	54 PD-N (63.4) 11 PD-MCI (69.5)	10-point clock drawing, frontal assessment battery	VBM	the thalamus, caudate and accumbens
Chung, et al. (Chung et al., 2017)	74 PD MCI converters (71.4) 108 PD-MCI nonconverters (69.2)	Semantic generative naming, Phonemic generative naming, Clock Drawing Test	ICV	the bilateral thalamus, right caudate, and right hippocampus
Crowley, et al. (Crowley et al., 2021)	25 PD Executive (69.1) 35 PD Memory (66.5) 56 PD Cognitively Well (68.8) 65 Non-PD (68.2)	D-KEFS - Tower Test (total achievement score)	PCA,DTI, ROI	less subcortical grey matter, white matter, and fewer bilateral dorsolateral-prefrontal cortex to caudate nucleus connections;
Brown, et al. (Brown et al., 2023)	12 PD (64.2) 12 HCs (63.3)	DRS2-attention subscore; Stroop color-word test, a Letter-Number sequence task	VBM, DTI	frontostriatal structural metrics
Study	Participants (mean age)	Assessment of executive function	MRI analysis	Correlation with brain structure
DTI				
Melzer, et al. (Melzer et al., 2013)	63 PD-N (64) 28 PD-MCI (71) 18 PD-D (73.7) 32 HCs (70.1)	Not mentioned	TBSS	prefrontal white matter, genu and internal and external capsules
Zheng, et al. (Zheng et al., 2014)	PD (n = 16) 62.2 who had undergone evaluation for stereotactic implantation of deep brain stimulators	Color-word interference, Letter fluency, Category fluency Category switching, Number-letter switching	ROI	frontal projection(ACR, ALIC, SFO)
Gallagher, et al. (Gallagher et al., 2015)	15 PD (60.2) 15 HCs (62.7)	TMT part B minus part A time difference, WCST-64 perseverative responses score, phonemic fluency/Controlled Oral Word Association Test score, the Stroop task interference score	TBSS	frontal white matter
Study	Participants (mean age)	Assessment of executive function	MRI analysis	Correlation with brain structure
Alzaid,et al. (Alzaid et al., 2020)	50 PD (66) 24 HCs (68)	TMT part A, B and B-A	TBSS	frontoparietal white matter alterations
Yang,et al. (Yang et al., 2021a)	19 Tremor-Dominant (TD) (57.3) 34 PD-N (57.5) 15 PIGD (57.8)	TMT A, B, Stroop Word Reading, Stroop Color Naming, Stroop Color-Word Interference; FAS Verbal Fluency, Animal Verbal Fluency	Probabilistic tractography	fronto-insular-striatal tracts
Scamarcia,et al. (Scamarcia et al., 2022)	154 PD (61.6) 60 HCs (61.8)	Digit span backward, Intra/Extra Dimensional Set Shift test (IED) from the CANTAB, and the Stroop color-word test	TBSS	WHM volume
rs-fMRI				
Baggio,et al. (Baggio et al., 2015b)	43 PD Non-MCI(64) 22 PD-MCI (66.1) 36 HCs (63.4)	Backward minus forward digit spans, TMT part A minus part B scores, phonemic fluency scores and Stroop Color-Word Test interference scores	rs-fMRI	reduced connectivity between the the dorsal attention network and right frontoinsula regions in MCI patients

(continued on next page)

Table 4 (continued)

Study	Participants (mean age)	Assessment of executive function	MRI analysis	Correlation with brain structure
Study	Participants (mean age)	Assessment of executive function	MRI analysis	Correlation with brain structure
Chantal et al. (Putcha et al., 2015)	24 PD (62.5) 20 MC (65.9)	MMSE	rs-fMRI	dysfunctional pattern of interaction between core neurocognitive networks
Chantal et al. (Putcha et al., 2016)	24 PD (62.5) 20 HC (65.9)	Trail Making A and B tests	rs-fMRI	dysfunctional coupling between the salience and default mode networks
Lang, et al. (Lang et al., 2019)	82 PD (70.5) 37 HCs (69.6)	DRS-2: Initiation/Perseveration subset, Letter Fluency (FAS), WAIS-IV Digit Span (Sequencing, BWD), STROOP Color-Word, Brixton Spatial Anticipation, Hayling Sentence Completion, TMT—B, Clock Drawing Test (Command)	rs-fMRI	degeneration in the sensorimotor network
Fathy, et al. (Fathy et al., 2020)	53 PD (67.3) 15 Controls (66.9)	Semantic Fluency, Intra-extra dimension shift	rs-fMRI	disconnection of the dAI with ACC
Study	Participants (mean age)	Assessment of executive function	MRI analysis	Correlation with brain structure
Maiti, et al. (Maiti et al., 2020)	81 PD (65.6) 43 Controls (62.4)	Verbal Fluency, Stroop and TMT	rs-fMRI	weaker FC between the vermis and dorsolateral prefrontal cortex
Devignes, et al. (Devignes et al., 2022)	31 PD-N (63.3) 14 PD-FS (67.3) 20 PD-PC (63.3) 30 PD-MS (67.2)	TMT-B, Stroop: interference time, Stroop: error Phonemic fluency: naming in 60 s, Alternating fluency: naming in 60 s, HVLIT frequency of retrieval deficit (%)	rs-fMRI	frontostriatal deficits had reduced inter-network FC between the visual, default-mode, sensorimotor, salience, dorsal attentional, basal ganglia, and frontoparietal networks
Di Tella, et al. (Di Tella et al., 2023)	26 PD (65.2) 24 HC (64.8)	Phonological Fluency, Semantic Fluency, TMT-A, TMT—B, TMTB-A	rs-fMRI	abnormal FC in executive-attentional networks
task-fMRI Tinaz et al. (Tinaz et al., 2008)	13 PD (57.6) 13 HC (57)	Digit Symbol, TMT-A, TMT—B, Digit Span, FAS letter fluency	task-fMRI	abnormal activation patterns of the prefrontal-striatal system
Study	Participants (mean age)	Assessment of executive function	MRI analysis	Correlation with brain structure
Ekman et al. 2012 (Ekman et al., 2012)	30 PD-MCI (67.8) 26 PD-without MCI (68.3) 24 HC (67.9)	WCST, Fluency (animal), TMT—B, Digit span back	task-fMRI, Presynaptic dopamine SPECT	decreased BOLD signals in the right caudate and frontal cortex, impaired presynaptic function in the right caudate.
Schendan et al. (Schendan et al., 2013)	12 PD (59) 12 MC (58) 14 YC (30)	MMSE, Wechsler Adult Intelligence Scale, TMT-A, TMT-B	task-fMRI	abnormal activation patterns and interactions within the prefrontal-striatal and medial temporal lobe regions
Nombela, et al. (Nombela et al., 2014)	49 Control (63.8) 49 Parkinson's disease (65.3)	Tower of London	task-fMRI	the frontoparietal network
Yang, et al. (Yang et al., 2021b)	15 PD-MCI (69) 26 PD-N (67.5) 22 HC (64.3)	TMT-A, TMT—B, Stroop test	task-fMRI	the Attention Network Test Alterations

Abbreviation: PD: Parkinson's Disease; PD-MCI: Parkinson's Disease with mild cognitive impairment; PD-N: Parkinson's Disease with normal cognition; PD-D: Parkinson's Disease with dementia; PIGD: postural instability/gait difficulty; PD-FS, Parkinson's disease frontostriatal subtype; PD-PC, Parkinson's disease posterior cortical subtype; PD-MS, Parkinson's disease mixed subtype; HC: Healthy control; YC: Younger control participants, MC: Healthy control participants matched to patients with Parkinson's disease; WCST: Wisconsin Card Sorting Test; DKEFS: Delis-Kaplan Executive Function System; FAB: the Frontal Assessment Battery; TMT-A: Trail Making Test A; TMT—B: Trail Making Test B; COWAT: Controlled Oral Word Association Test; MoCA: Montreal Cognitive Assessment; DRS2: the Dementia Rating Scale 2.

DTI: diffusion tensor imaging; ROI: Region of Interest; VBM: Voxel-Based Morphometry; PCA: Principal component analysis; TBSS: Tract-based spatial statistics; rsfMRI: resting-state functional MRI analyses ; task-fMRI: task-functional MRI; FC: functional connectivity; ACR: Anterior corona radiata; ALIC: Anterior limb of internal capsule; SFO: Superior fronto-occipital fasciculus; dAI: dorsal anterior insula; ACC: anterior cingulate cortex
DMN: the default mode network; FPN: the frontoparietal network; RSNs: Resting state networks.

atrophy of the left middle temporal gyrus is associated with impairments in executive function, attention and memory, whereas atrophy of the superior frontal gyrus is only associated with attention deficits (Mak et al., 2014a). Christopher et al. observed significant grey matter loss in the limbic/paralimbic systems in PD patients compared to controls, and the impairments in executive and visuospatial functions were correlated with grey matter loss in the dorsolateral prefrontal cortex and parahippocampal gyrus in PD patients (Christopher and Strafella, 2013).

Extensive alterations in subcortical structures also contribute to the development of EDF in PD patients (Gao et al., 2017). The cortico-

striatal circuitry is considered the primary anatomical basis for anatomical underpinning of many symptoms in PD, including EDF. One study demonstrated that patients with Parkinson's disease with mild cognitive impairment (PD-MCI) exhibited significant volume reductions in multiple subcortical grey matter regions compared to patients with Parkinson's disease with normal cognition (PD-N), including the thalamus and nucleus accumbens, with a trend towards the caudate nucleus (Mak et al., 2014a). Foo et al. also found that individuals who transitioned from PD-N to PD-MCI exhibited more pronounced nucleus accumbens atrophy at baseline, and progressive atrophy of the

thalamus, caudate nucleus, and nucleus accumbens was associated with memory and executive function impairments (Foo et al., 2017). Callaghan et al. observed a direct correlation between EDF and the dorsal striatum (particularly the head of the dorsolateral caudate nucleus), attributing this phenomenon to its close connection with the lateral prefrontal cortex (O'Callaghan et al., 2014). Anderkova et al. concluded that the connectivity between the mid-striatum and the precuneus was negatively associated with executive function (Anderkova et al., 2017). Beyond the striatum, the insular cortex has been implicated in a variety of different cognitive processing tasks; Mak et al. proposed that insular atrophy is related to impairments in executive function and attention (Mak et al., 2014b). The insula and the basal ganglia circuitry, as well as many other cortical areas including the frontal, temporal, parietal and cingulate cortices, are closely related and can interact with multiple brain network regions, involving a multi-faceted approach to cognition, emotion, sensory and autonomic processes.

In summary, the reduction in grey matter volume, especially in the frontal lobes and subcortical grey matter, constitutes the basis of frontal-striatal pathway dysfunction, which lays the foundation for cognitive impairments in PD (Wen et al., 2015). The anterior insula is connected to the anterior cingulate cortex, thus facilitating the network transitions required for executive functions (Christopher et al., 2014a).

7.2. Diffusion Tensor Imaging (DTI)

Diffusion tensor imaging (DTI) is commonly used to investigate changes in white matter. Alterations in DTI indices may result from neuronal loss and gliosis, as well as dysregulation of axonal membranes, myelin, microtubules, and neurofilaments (Hirano, 2021). Decreased fractional anisotropy (FA) or increased mean diffusivity (MD) and radial diffusivity (RD) suggest impaired white matter integrity (Uluğ et al., 1999). Parkinson's disease patients exhibit widespread pathological changes in white matter, which are associated with cognitive impairments (Rae et al., 2012). Grey et al. included 11 articles investigating the correlation between white matter and cognitive function in PD and conducted a meta-analysis, concluding that eight studies support the view that white matter damage caused by cerebrovascular disease negatively affects cognitive function in PD patients, while three studies did not report this (Grey et al., 2019). Scamarcia et al. proposed that the volume of white matter hyperintensities changes over time and is associated with global cognition, executive function, and language impairment. Baseline white matter hyperintensity volume is a moderate risk factor for progression to mild cognitive impairment (Scamarcia et al., 2022). The corpus callosum, a key commissural fiber connecting the bilateral cerebral hemispheres, may contribute to PD cognitive impairments by disrupting information transfer between hemispheres and from the corpus callosum to the cortex (Bledsoe et al., 2018). Studies have shown that localized atrophy in different parts of the corpus callosum can predict specific cognitive deficits, with attention/working memory linked to central body volume, executive function, language and memory associated with mid-posterior body volume, and memory and visuospatial function related to posterior body volume (Goldman et al., 2017). Multiple studies have found that executive dysfunction in PD is largely associated with damage to frontal lobe white matter and projecting fibers (Zheng et al., 2014; Brown et al., 2023; Melzer et al., 2013). Koshimori et al. using tract-based spatial statistics (TBSS), showed that PD patients exhibit more extensive white matter damage primarily in bilateral frontal and temporal regions. This study further found that white matter damage in the fronto-temporal region was closely related to overall cognitive function and executive function. This result, combined with previous studies, suggests that frontal lobe white matter damage is a core pathological basis for mild cognitive impairments in PD (Koshimori et al., 2015).

The degeneration of white matter in PD patients can lead to EDF through two main factors: on one hand, damage to the deep white matter of the frontal lobe may cause dysfunction of the frontal-striatal

circuit, thus forming the basis for executive dysfunction (Crowley et al., 2021); on the other hand, the specific mechanisms of cognitive dysfunction in Parkinson's disease induced by vascular pathology remain controversial and hypothetical. Currently, there are two possible mechanisms: the first suggests that cognitive dysfunction arises from two independent but additive disease processes, neurodegenerative disease and cerebrovascular pathology. However, many theories do not support this view, arguing that most PD patients do not have significant cerebrovascular pathology. The second, more probable mechanism involves the synergistic impact of these two conditions, where subclinical perfusion deficits adversely affect vulnerable regions through the neurodegenerative process. Regions with subclinical hypoperfusion are predisposed to enter a degenerative process, thereby incurring damage (Blume et al., 2017).

7.3. Functional Magnetic Resonance Imaging (fMRI)

fMRI imaging techniques rely on local cerebral oxygen content changes and are thus known as blood oxygen level dependent (BOLD) brain function imaging, a technique based on the magnetic susceptibility changes around blood vessels induced by variations in the concentration of deoxyhemoglobin. Currently, two main brain network research methods are used in fMRI: resting-state fMRI (rs-fMRI) and task-related fMRI.

Resting-state fMRI can detect individual resting-state low-frequency spontaneous fluctuations in the BOLD signal and can be used to study spatially distributed brain networks' intra- and inter-network functional connectivity (Ryman and Poston, 2020).

The default mode network (DMN) dysfunction can appear early in the disease course of PD patients, during the mild cognitive impairment stage, or even in the absence of cognitive impairment. The DMN is a brain network that is activated during rest or passive visual fixation and deactivates (becomes less active) during cognitive tasks, meaning it is more excited during rest or baseline states than during active or goal-oriented processing of environmental stimuli (Tessitore et al., 2012). This network mainly includes the precuneus/posterior cingulate cortex, medial prefrontal cortex, inferior parietal lobule, and posterior medial and lateral temporal cortical regions (Prodoehl et al., 2014; Baggio et al., 2015a). These brain areas have important physiological functions and are related to processes such as episodic memory, self-reflection, and cognitive processing (Tahmasian et al., 2015). Connectivity between the striatum and frontal cortex is related to executive function, a finding that has been confirmed in the early stages of PD. It can be hypothesized that connectivity deficits in this region may be related to Parkinson's-specific executive function deficits. Similarly, the precuneus, as the hub of the DMN (with reduced functional connectivity to the prefrontal cortex), and the reduced functional connectivity of other DMN nodes, are associated with the cognitive dysfunctions observed in PD patients (Montgomery Jr., 2015; Wolters et al., 2019). Chantal Stern's research proposed that patients with PD exhibit coupling disorders among three key neural networks: the salience network (SN), the central executive network (CEN), and the default mode network (DMN). These disorders are associated with cognitive impairments, which may manifest as symptoms including distracted attention, difficulty in decision-making, and executive dysfunction (Putcha et al., 2015). Further research has suggested that the functional coupling disorder between the SN and DMN in PD patients contributes to attention deficits during high-demand cognitive tasks, potentially leading to the development of dementia (Putcha et al., 2016).

The frontoparietal network (FPN) includes the lateral prefrontal cortex and the posterior parietal cortex, and is known to be involved in various cognitive tasks by initiating and regulating cognitive control capacities. Depending on the nature of the task, the FPN can flexibly connect with the DMN or the dorsal attention network (DAN) and it is thought to regulate the dynamic balance between these neural networks (Zanto and Gazzaley, 2013). Studies indicate that reduced functional

connectivity in the FPN may be linked to the development of mild cognitive impairment in PD patients. These findings are consistent with previous task-related fMRI studies that have emphasized a reduction in frontal-striatal excitability in PD patients with cognitive deficits, as well as results from rs-fMRI studies in PD-D. Furthermore, a recent rs-fMRI study investigating connections between internal brain networks has highlighted reduced connectivity between the DMN, DAN and the FPN-DAN networks, which is associated with cognitive decline in PD patients. Importantly, the identification of disruptions in DMN and FPN connectivity in PD-MCI patients without significant grey matter (GM) changes suggests that functional alterations may precede structural abnormalities in GM (Amboni et al., 2015).

In human and primate brains, there are two attention networks: the DAN and the ventral attention network (VAN). The key components of the DAN include the bilateral intraparietal sulcus and the frontal eye fields (FEF), which are related to top-down, goal-directed attention. The VAN comprises the temporoparietal junction (TPJ) and ventral prefrontal cortex (VFC), and it responds to stimuli from the environment (bottom-up attention) (Farrant and Uddin, 2016; Baggio et al., 2015b). An rs-fMRI study exploring brain network alterations in cognitive impairment in Parkinson's disease found that, according to independent component analysis (ICA) results, connectivity between the DAN and the right frontal-insular region was reduced in Parkinson's patients with mild cognitive impairment, correlating with poorer attention/executive function (Baggio et al., 2015b).

Changes in cerebellar network connectivity also contribute to the development of EDF in PD patients. Research by Maiti et al. found that functional connectivity between the DLPFC and cerebellum was significantly reduced in PD patients with cognitive impairment compared to those without cognitive deficits. Reduced connectivity in the cerebellar vermis fibers was related to deficits in visuospatial abilities, attention and executive functions (Maiti et al., 2020; Shuai et al., 2020). Shuai et al. also observed a reduction in cerebellar network connectivity, speculating that this may be due to disruptions in the basal ganglia-thalamo-cortical circuit, leading to decreased functional connectivity between the cerebellum and caudate network, as well as between the cerebellum and the insular network. Their study also suggested that in the early stages of cognitive impairment in PD patients, there is an initial short-term increase in functional connectivity within the executive network, followed by executive dysfunction (Shuai et al., 2020). Other research posits that degeneration of the sensorimotor network leads to EDF; while decreased connectivity between the temporal lobes and the central executive network is associated with dysfunction in posterior cortical regions (Lang et al., 2019).

Task-related fMRI approaches have explored the possible neural network mechanisms underlying EDF in PD, finding that striatal dopamine depletion impairs the information processing capacity of the cortical-striatal circuit. Specifically, research has shown that PD-MCI exhibit decreased BOLD signals in the right caudate nucleus and frontal cortex during working memory tasks, compared to patients without MCI, alongside presynaptic dysfunction in the right caudate nucleus (Ekman et al., 2012). These results indicate that PD executive dysfunction is due to a reduction in striatal dopamine, leading to decreased excitability and connectivity among task-related brain networks. Initially, patients may compensate for the decline in network connectivity and efficiency through increased excitability. When such over-excitation is no longer capable or sufficient to compensate for reduced connectivity, cognitive impairments become apparent in behavior. Studies have also found, in line with previous results, that dopamine supplementation can improve working memory and certain cognitive functions. Hence, it is hypothesized that dopamine supplementation may restore network connectivity and thereby cognitive performance.

7.4. Positron emission tomography/computed tomography (PET/CT)

PET brain imaging plays a crucial role in visualizing the distribution

of radiotracers in vivo, enabling the detection of characteristic pathophysiological and neurochemical changes in PD (Thobois et al., 2001). It has become an essential tool for early diagnosis, differential diagnosis and the study of pathophysiological mechanisms in Parkinson's disease (Sarikaya, 2015). The most commonly used PET imaging in PD clinical practice includes dopaminergic neuron imaging (presynaptic membrane, synaptic cleft, postsynaptic membrane imaging) and glucose metabolism imaging. Additionally, amyloid β ($A\beta$) imaging, Tau imaging and neuroinflammation imaging are also being researched and discussed (Table 5).

7.5. Dopaminergic neuron imaging

7.5.1. Presynaptic membrane

Chung et al. through 18F-FP-CIT PET-CT imaging methods, divided the striatum into four sub-regions and concluded that caudate nucleus dopamine depletion mediates and directly or indirectly affects frontal/executive functions (Chung et al., 2022). Kim et al. studied the relationship between striatal dopamine transporter (DAT) loss, cognitive function and cortical metabolism in PD patients. Apart from the posterior putamen, striatal DAT loss was associated with cognitive impairment in PD patients, particularly frontal/executive and visuospatial functions (Kim et al., 2019). Furthermore, the 6-[18F]-fluoro-L-DOPA (FDOPA) PET scan has been widespread in quantifying dopamine metabolism in the brain, establishing the pathogenic role of striatal dopaminergic depletion in the early impairment of executive function in PD and confirming that striatal dopamine depletion leads to executive dysfunction in PD patients (Cheesman et al., 2005). Gallagher et al. found that subtle changes in dopamine metabolism in the anterior cingulate gyrus may lead to executive dysfunction behaviors in Parkinson's disease (Gallagher et al., 2015).

7.5.2. Postsynaptic membrane

Idiopathic Parkinson's disease is often associated with executive function deficits similar to those seen in patients with frontal lobe lesions. Christopher et al. explored changes in synaptic dopamine levels in PD patients during the execution of spatial working memory tasks (SWT), monitored using the D2 receptor radioligand 11C-raclopride (RAC) PET. Statistical analyses comparing control and PD patient groups showed a significant decrease in dopamine release in the dorsal caudate nucleus of PD patients, while dopamine release levels in the medial prefrontal cortex remained relatively stable. These results support the hypothesis that striatal dopamine depletion is involved in executive function deficits in PD patients with mild symptoms. Early deficits in executive function are linked to impaired nigrostriatal dopamine function, affecting processing in the cortico-basal ganglia circuit. Conversely, mesocortical dopaminergic transmission in early PD patients appears to be relatively preserved (Sawamoto et al., 2008). Therefore, dopamine replacement therapy may be beneficial for the striatum but detrimental to the frontal cortex, as overstimulation of dopamine receptors in the frontal cortex may affect cortical function (Sawamoto et al., 2007). Another study examined the role of the insula in the cognitive decline of early Parkinson's disease (PD) patients. The study utilized 11C-dihydrotetrabenazine (DTBZ) and 11C-FLB 457 to evaluate the availability of cortical D2 receptors in PD-N, PD-MCI and healthy controls (HCs) matched by sex and age. The findings revealed that PD patients with mild cognitive impairments showed significant striatal dopamine depletion in associative subregions and reduced availability of D2 receptors in the bilateral insula, even after adjusting for age, disease severity and dopaminergic drug intake. The study also highlighted the relationship between striatal dopamine depletion and loss of D2 receptors in the insula among PD patients with mild cognitive impairments, emphasizing the importance of dopaminergic neurons and the interplay between striatal and insular dopamine in cognitive decline. The function of dopaminergic neurons and the relationship between striatal and insular dopamine may be important contributors to

Table 5

Summary of main PET/SPECT studies on neuroanatomical correlates of Executive dysfunctions in PD.

Study	Participants (mean age)	Assessment of executive function	PET/SPECT radiotracer	Correlation with brain structure
PET				
Marié, et al. (Marié et al., 1999)	10 PD non-demented(59.9)	Object alternation Conditional associative learning Brown-Peterson paradigm	11C-S-NMF	striatum
Cheesman, et al. (Cheesman et al., 2005)	16 PD non-demented, non-depressed (61.3)	the one-touch Tower of London (TOL) task Verbal Phonological Fluency (letter cues: F, A, and S) Verbal Semantic Fluency (category: animals)	[18F]FDOPA	striatum
Sawamoto, et al. (Sawamoto et al., 2008)	7 PD (53.9) 7 HCs (51.4)	a spatial working memory task	11C-raclopride	striatum
Study	Participants (mean age)	Assessment of executive function	PET/SPECT radiotracer	Correlation with brain structure
Cropley, et al. (Cropley et al., 2008)	15 PD (62.1) 14 Controls (61.6)	a computerized version of the WCST and the Stockings of Cambridge task from the Cambridge Automated Neuropsychological Test Battery (Cambridge Cognition, UK)	[11C]NNC 112 ; [18F] FDOPA	striatum
Christopher, et al. (Christopher et al., 2014b)	11 PD-N (68.8) 11 PD-MCI (70.8) 14 HCs (67.5)	Color-Word Interference (Stroop task), inhibition (executive); Verbal Fluency, total score for category fluency	11C-FLB 457 11-Cdihydrotetabenazine	insula
Gallagher, et al. (Gallagher et al., 2015)	15 Non-demented early-stage PD patients (60.3)	Stroop Color-Word score, TMT B-A, WCST-64 Perseverative Responses score	[18F]FDOPA	anterior cingulate
Meles, et al. (Meles et al., 2015)	16 PD (63.7) 17 HCs (61.1)	FAB	FDG PET	PDCP
Mattis, et al. (Mattis et al., 2016)	89 PD (61.6)	the Stroop Tests, WCST, Symbol Digit Modality Test (SDMT),and TMT	18F-FDG	PDCP
Study	Participants (mean age)	Assessment of executive function	PET/SPECT radiotracer	Correlation with brain structure
Wu, et al. (Wu et al., 2018)	168 PD (61.5) 100 HCs (62.1)	the Stroop Color-Word Test part C, TMT part B	18F-FDG	cuneus, superior temporal gyrus, and inferior parietal lobule
Kim, et al. (Kim et al., 2019)	13 PD (62.2) 15 PD-MCI (63.9)	COWAT, Korean-Color Word Stroop Test (K-CWST)	FP-CIT FDG	DAT loss in the striatum, except in the posterior putamen
Apostolova, et al. (Apostolova et al., 2020)	41 PD (61.4)	not mentioned	FP-CIT FDG	the cognitive striatum; medial parts of the frontal cortex (ACC included)
Chung, et al. (Chung et al., 2022)	240 PD (70.6)	COWAT animal; COWAT supermarket, COWAT phonemic, and the Stroop color reading test	18F-FP-CIT DTI	caudate
Han, et al. (Han et al., 2021)	21 PD-N (60.4) 20 PD-MCI (62.4)	Stroop Color-Word Test (CWT) and TMT-B	11C-CFT 18F-FDG	striatum, the inferior frontal gyrus, putamen, and insula.
Study	Participants (mean age)	Assessment of executive function	PET/SPECT radiotracer	Correlation with brain structure
Lorek, et al. (Lorek et al., 2023)	11 PD (65.4)	TMT-A, TMT—B, Stroop test, parts I and II	[18F] FDOPA	striatal dopaminergic deficiency
Zhihui, et al. (Zhihui et al., 2023)	84 PD (63.5)	TMT-B, Draw clock, Semantic fluency, Abstraction	18F-FDG	left lateral prefrontal cortex
Schröter, et al. (Schröter et al., 2023)	17 PD-DEM (67.1) 23 PD-N (60.5) 36 PD-MCI (63.6)	Category fluency	FDG	PDCP
Yacoubian, et al. (Yacoubian et al., 2023)	58 PD (66.1) 62 Controls (64.1)	DKEFS Color/Word Interference, TMT	[18F] DPA-714	putamen, thalamus, substantia nigra, and cortical brain regions
Ghadery, et al. (Ghadery et al., 2020)	17 PD (63.9) 12 HCs (64.0)	MoCA	[18F]-FEPPA	DLPPFC, VLPPFC and MPFC
SPECT				
Ekman, et al. (Ekman et al., 2012)	77 PD (67.6) 24 Controls (67.9)	WCST, Fluency (animal), TMT—B, Digit span back	123I-FP-CIT	caudate
Study	Participants (mean age)	Assessment of executive function	PET/SPECT radiotracer	Correlation with brain structure
Pasquini et al. (Pasquini et al., 2019)	397 PD (61.7) 177 HCs (61)	MoCA	123I-FP-CIT	bilateral caudate
Stögbauer, et al. (Stögbauer et al., 2020)	54 PD (68)	TMT-A, TMT—B, Phonemic Fluency	123I-FP-CIT	striatum

Abbreviation: PD: Parkinson's Disease; HC: Healthy control; PD-MCI: Parkinson's Disease with mild cognitive impairment; PD-N: Parkinson's Disease with normal cognition; PD-DEM: patients classified as suffering from dementia according to DRS-2 < 133, DRS-2, Mattis Dementia Rating Scale 2;TOL: Tower of London; WCST: Wisconsin Card Sorting Test; DKEFS: Delis-Kaplan Executive Function System; COWAT: Controlled Oral Word Association Test; TMT—B: Trail Making Test B; TMT-A:

Trail-making Test A; MoCA: Montreal Cognitive Assessment; PD: Parkinson's Disease; PDCP: Parkinson's Disease related cognitive pattern; DLPFC: Dorsolateral prefrontal cortex, VLPFC: Ventrolateral prefrontal cortex; MPFC: Medial prefrontal cortex.

cognitive decline. These findings suggest that striatal dopaminergic denervation combined with insular D2 receptor deficits underlie mild cognitive impairments in PD, particularly the decline in executive function. Moreover, these studies indicate that the insula, modulated by dopamine, plays a key and direct role in facilitating cognitive functions (Christopher et al., 2014b).

7.6. Glucose metabolism imaging

Lorek et al. utilized 18F-FDG PET measurements to investigate cortical dysfunction and degeneration, leading to the conclusion that nigrostriatal degeneration plays a role in the pathogenesis of cognitive deficits associated with medial frontal lobe dysfunction, such as impaired inhibitory control (Lorek et al., 2023). In 2008, Huang et al. proposed the PD-related cognitive pattern (PDCP), characterized by reduced metabolism in frontal and parietal regions and a relative increase in the cerebellar vermis and dentate nucleus, using 18F-fluorodeoxyglucose (FDG) PET imaging. This pattern's expression was correlated with neuropsychological tests of memory and executive function in a cohort of PD patients undergoing prospective assessment and could be used for monitoring and assessing the progression of cognitive functions in patients, with the potential to become a biomarker for cognitive impairments in PD (Huang et al., 2008). Subsequent studies have consistently confirmed that the expression of PDCP increases with the decline in cognitive and executive functions (Schröter et al., 2023; Meles et al., 2015; Mattis et al., 2016).

7.7. Neuroinflammatory imaging

Neuroinflammation, characterized by microglial activation, is a key factor in Parkinson's disease. Ghadery et al. conducted a dual-tracer PET study examining the relationship between microglial activation transporter protein (translocator protein, TSPO) 18-kDa labeled with the radioactive ligand [18F]-FEPPA and β -amyloid (labeled with [11C] PIB). The results indicated a significant interaction between β -amyloid and neuroinflammation, particularly in patients with cognitive deficits, localized in the striatum and frontal lobe regions (i.e. DLPFC, VLPFC and MPFC), affecting executive function and decision-making processes in individuals with Parkinson's disease (Ghadery et al., 2020).

7.8. Single-photon emission computed tomography (SPECT)

Multiple SPECT studies suggest that executive dysfunction in PD is attributed to dysfunction in the frontal-striatal brain circuit. One study using PET or SPECT imaging found that reduced dopamine transporter uptake in the caudate nucleus of PD-D patients was associated with executive function impairment (Aarsland et al., 2017). Another study assessed brain responses during working memory tasks in newly diagnosed, drug-naïve PD patients with and without mild cognitive impairment (MCI). The integrity of presynaptic dopamine was measured using ^{123}I -FP-CIT SPECT. The results indicated that PD patients with MCI exhibited lower presynaptic ^{123}I -FP-CIT binding in the right caudate nucleus compared to those without MCI. The caudate nucleus, being crucial for working memory, is implicated in the impairment of working memory performance and EDF in PD patients, potentially due to the reduced integrity of dopamine in the dorsal caudate nucleus, which is a primary output target of the nigrostriatal pathway. Abnormal regulation of the nigrostriatal dopaminergic circuit is related to cognitive impairments in early PD patients (Ekman et al., 2012).

8. Conclusion and outlook

Executive function is the basis for many cognitive, affective and

social skills and is among the most complex behaviors. Impairment in executive function can occur early in PD patients. This review discusses the characteristics and potential mechanisms of EDF in PD from multiple perspectives, including the concept of executive function, commonly used neuropsychological scales, neurotransmitters, genetics, electroencephalographic activity and neuroimaging. Degeneration of the frontal-striatal circuit, as well as mutations in the *COMT* and *LRRK2* genes, are closely related to executive dysfunction in PD patients. Cholinergic neurotransmitter pathways in the posterior cortical regions and mutations in the *MAPT* and *GBA* genes are closely related to the progression of PD patients to dementia. The observed increase in theta and delta wave activity, alongside a decrease in alpha wave activity, may represent promising biomarkers for the early detection and monitoring of EDF, as well as the progression of dementia in patients with PD. The PDCP pattern can be used for monitoring and assessing the progression of cognitive functions in patients and is expected to become a biomarker for cognitive disorders in PD. In the future, integrating neuropsychological scales, multifunctional neuroimaging, genetics and neurobiochemistry may provide a theoretical basis for more precise diagnosis of executive dysfunction in PD. There are currently no effective medications specific for the improvement of EDF in PD patients. Targeted treatments based on the imaging and genetic phenotypes of PD patients may better delay the progression of cognitive impairments.

Compliance with ethics requirements

Our article does not contain any studies with human or animal subjects.

CRediT authorship contribution statement

Shuyan Tong: Writing – review & editing, Writing – original draft, Formal analysis, Data curation, Conceptualization. **Ruiwen Wang:** Writing – review & editing, Writing – original draft, Conceptualization. **Huihua Li:** Writing – review & editing. **Zhu Tong:** Writing – review & editing. **Deqin Geng:** Writing – review & editing. **Xiangrong Zhang:** Writing – review & editing, Validation, Supervision, Investigation, Conceptualization. **Chao Ren:** Writing – review & editing, Validation, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

We declare there is no conflict of interest.

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Data availability

No data was used for the research described in the article.

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